

Expert System written in CLIPS

Lum Meta - ICT – AI

In this project we aim to create an expert system using clips to determine if a patient has diabetes or not

We can start by analyzing the rules we need to add

1. Blurred vision (Boolean: Yes/No)
2. Tiredness (Boolean: Yes/No)
3. Body Mass Index (BMI) > 28
4. Age > 25
5. familial history (Boolean: Yes/No)
6. Fasting blood sugar (FBS) >= 126 mg/dL
7. Low physical activity i.e. less than 3 times per week (Boolean: Yes/No)
8. Numbers of Pregnancies >= 3

Now we can create a template that matches and has all the information needed, and this is the command for that:

```
(deftemplate patient
(slot name)
(slot weight)
(slot height)
(slot blurred-vision)
(slot tiredness)
(slot BMI)
(slot age)
(slot family-history)
(slot fasting-blood-sugar)
(slot low-physical-activity)
(slot number-of-pregnancies))
```

And we can confirm that the template was added by using (get-deftemplate-list)

```
CLIPS> (get-deftemplate-list)
(patient)
CLIPS>
```

Because the BMI need to be calculated let's make a rule that does that, for this to work initially the patient's BMI needs to be entered as 0.0 and when the rule detects this it will calculate the BMI and update it

And this is the command for that rule:

```
(defrule calculate-BMI(patient (name ?n)
(weight ?w)
(height ?h)
(BMI ?b&:(= ?b 0.0)))
=> (bind ?calculated-bmi (/ ?w (* ?h ?h)))
(modify 1 (BMI ?calculated-bmi))
(printout t "Calculated BMI for " ?n ": " ?calculated-bmi crlf))
// The print line will be removed after testing
```

Lets now test this rule with this test info:

```
(assert (patient (name "Alice")
(weight 70)
(height 1.65)
(blurred-vision yes)
(Tiredness no)
(BMI 0.0)
(age 30)
(family-history yes)
(fasting-blood-sugar 130))
```

```
(low-physical-activity yes)
(number-of-pregnancies 3)))
```

Running the agenda command we can see that the calculate-BMI rule will trigger and change the BMI of the fact

```
CLIPS> (run)
```

```
Calculated BMI for Alice: 25.7116620752984
```

```
CLIPS> (facts)
```

```
f-1  (patient (name "Alice") (weight 70) (height 1.65) (blurred-vision yes)
      (Tiredness no) (BMI 25.7116620752984) (age 30) (family-history yes) (fasting-
      blood-sugar 130) (low-physical-activity yes) (number-of-pregnancies 3))
```

Running the facts command we can see that the BMI was updated, and now the rule can be updated to remove the printout line.

Now that we have the BMI, we can create the rule for diabetes diagnosis:

```
(defrule dignosis (patient (name ?n)
  (blurred-vision yes)
  (tiredness yes)
  (BMI ?b&:(> ?b 28))
  (age ?a&:(> ?a 25))
  (family-history yes)
  (fasting-blood-sugar ?fbs&:(>= ?fbs 126))
  (low-physical-activity yes)
  (number-of-pregnancies ?preg&:(>= ?preg 3)))
=>
(printout t "Patient " ?n " meets ALL conditions: TRUE" crlf))
```

lets try if this rule work, I also added another patient that fits all of the rules for diagnosis, and these are the results:

f-1 (patient (name "Alice") (weight 72) (height 1.65) (blurred-vision yes) (tiredness no) (BMI 101.851851851852) (age 30) (family-history yes) (fasting-blood-sugar 130) (low-physical-activity yes) (number-of-pregnancies 3))

f-2 (patient (name "Jack") (weight 330) (height 1.8) (blurred-vision yes) (tiredness yes) (BMI 0.0) (age 30) (family-history yes) (fasting-blood-sugar 130) (low-physical-activity yes) (number-of-pregnancies 3))

As we can see the rule is not updating the BMI for fact number 2 (Jack), this is happening because there is a mistake on the rule for calculating the BMI

(modify 1 (BMI ?calculated-bmi)) this is the mistake, “modify 1” only changes the fact with the index 1

```
(defrule calculate-BMI ?f <- (patient (name ?n)
(weight ?w)
(height ?h)
(BMI ?b&:(= ?b 0.0)))
=>
(bind ?calculated-bmi (/ ?w (* ?h ?h)))
(modify ?f (BMI ?calculated-bmi)))
```

This is the new rule code, this way we always update the fact that was last checked.

When running the new system, we get this output:

Patient Jack meets ALL conditions: TRUE

This means that the BMI is being updated for all the patients that's proved when we check all the facts

f-1 (patient (name "Alice") (weight 72) (height 1.65) (blurred-vision yes) (tiredness no) (BMI 26.4462809917355) (age 30) (family-history yes) (fasting-blood-sugar 130) (low-physical-activity yes) (number-of-pregnancies 3))

f-2 (patient (name "Jack") (weight 330) (height 1.8) (blurred-vision yes) (tiredness yes) (BMI 101.851851851852) (age 30) (family-history yes) (fasting-blood-sugar 130) (low-physical-activity yes) (number-of-pregnancies 3))

When running "(agenda)" command we get this result:

```
0 calculate-BMI: f-2
0 calculate-BMI: f-1
```

At first it seems like the diagnosis rule is never triggered, but that's because the BMI for all patients is 0.0 before it is calculated so it is not recognized that the rule will trigger. When we use "(watch activations)" we will see that the "calculate-BMI" rule is triggered after each new patient is added :

```
==> Activation 0 calculate-BMI: f-1
==> Activation 0 calculate-BMI: f-2
```

And then we use the "run" command we get this output:

```
==> Activation 0 diagnosis: f-2
```

Meaning that the "diagnosis" rule will be triggered by fact 2 after the BMI was updated

I now added 10 patients and this is the result from running the system:

```
==> Activation 0 diagnosis: f-8
Patient Amira meets ALL conditions: TRUE
==> Activation 0 diagnosis: f-5
Patient Sam meets ALL conditions: TRUE
==> Activation 0 diagnosis: f-3
Patient Carlos meets ALL conditions: TRUE
==> Activation 0 diagnosis: f-1
Patient Jack meets ALL conditions: TRUE
```

And all the patients are diagnosed correctly, but the issue here is that the system requires 3 pregnancies to diagnose the patients with diabetes, but male patients will always have a pregnancy count of 0 so the system will never diagnose them with diabetes.

In this case we can either remove the pregnancy count or ignore it when the patient is a male, but a better approach would be to count the number of symptoms and if the patient has more than 5 symptoms they are considered at high risk of diabetes.

For this we can use a variable to count the symptoms and if statements to increment the variable for each symptom the patient has and then check if the value is greater than the 5 to recommend visiting a doctor.

This is the rule for that:

```
(defrule diagnosis(patient (name ?n)
(age ?a)
(weight ?w)
(blurred-vision ?bv)
(tiredness ?t)
(BMI ?bmi)
(family-history ?fh)
(fasting-blood-sugar ?fbs)
(low-physical-activity ?lpa)
(number-of-pregnancies ?nop))
=>
(bind ?count 0)
(if (> ?a 25) then (bind ?count (+ ?count 1)))
(if (eq ?bv yes) then (bind ?count (+ ?count 1)))
(if (eq ?t yes) then (bind ?count (+ ?count 1)))
(if (> ?bmi 28) then (bind ?count (+ ?count 1)))
(if (eq ?fh yes) then (bind ?count (+ ?count 1)))
(if (>= ?fbs 126) then (bind ?count (+ ?count 1)))
(if (eq ?lpa yes) then (bind ?count (+ ?count 1)))
(if (>= ?nop 3) then (bind ?count (+ ?count 1)))
(assert (symptom-count ?count))
(if (> ?count 5) then
```

```
(printout t "Patient " ?n " has a high risk of diabetes (symptom count: " ?count
")." crlf))))
```

This uses the variable “count” to count the number of symptoms the patient has and if the count greater then 5 that patient is considered at high risk of having diabetes

Now let's test the system with a few patients at once:

Running the (agenda) command now we see that both rules are triggered for each patient:

```
0 calculate-BMI: f-4
0 diagnosis: f-4
0 calculate-BMI: f-3
0 diagnosis: f-3
0 calculate-BMI: f-2
0 diagnosis: f-2
0 calculate-BMI: f-1
0 diagnosis: f-1
...
```

And running the system this is the result:

Patient Amira has a high risk of diabetes (symptom count: 6).

Patient Nina has a high risk of diabetes (symptom count: 6).

Patient Sam has a high risk of diabetes (symptom count: 8).

Patient Carlos has a high risk of diabetes (symptom count: 8).

Patient Jack has a high risk of diabetes (symptom count: 7).

Now we can see that even the male patients are being considered as high risk patients, now the pregnancy count is not a problem.

